

Quinn Mair

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PROFESSIONAL SUMMARY

Electrical and Computer Engineering student at Duke University with experience spanning controls, industrial automation, product development, and technical leadership. Proven ability to translate complex technical requirements into practical solutions, communicate across technical and non-technical teams, and lead projects from concept through implementation. Interested in customer-facing systems and solutions engineering roles at the intersection of technology, problem-solving, and business.

EDUCATION

Duke University Pratt School of Engineering

BSE in Electrical and Computer Engineering, Minor in Cultural Anthropology.

Durham, North Carolina

Expected May 2027

- Relevant Coursework: Digital Systems, Signal Processing, Product Design, Entrepreneurship

George Washington High School

International Baccalaureate Diploma. Valedictorian.

Denver, Colorado

Graduated May 2023

EXPERIENCE

Automation Systems/EE Design Intern

June 2026 – August 2026

HDR

Denver, CO

- Drafted and reviewed instrumentation drawings and electrical one-line diagrams for water treatment plant control systems
- Developed an Excel calculator for a client to determine bus overhaul facility footprint size based on bus input parameters
- Programmed and validated PLC ladder logic using Do-more Designer for industrial automation applications
- Supported multidisciplinary design reviews involving controls, electrical, and instrumentation systems

On-Site Construction Manager

May 2025 – August 2025

Engineers In Action

Shiselweni, Eswatini

- Coordinated daily activities of 40+ workers to deliver the longest footbridge in Eswatini (128m)
- Communicated project progress and technical decisions in Siswati during construction and inauguration
- Drafted and iterated drawings using field data and AutoCAD
- First Duke student awarded Emerging Leader of the Year at the annual EIA conference

President

January 2026 – Present

Duke Engineers for International Development

Durham, NC

- Lead planning, budgeting, implementation, and documentation of 4 infrastructure projects across 3 continents
- Fundraise and allocate \$70,000+ annual budget; personally raised ~\$100,000 over 3 years

Co-Founder

May 2024 – September 2025

Lunch Bunch LLC

Durham, NC

- Founded a lunch tray startup: researched and developed an interlocking lunch tray for local schools
- Drafted and submitted a nonprovisional patent, conducted prior art search, and responded to office actions
- Awarded CFCI GoBig Grant and Duke I&E support grant (x2); featured in Duke Startup Showcase (top 18 of 93)

Teaching Assistant, EGR 105

August 2025 – December 2025

Duke University

Durham, NC

- Instructed 200+ students in recursion, algorithm development, Python syntax, and computational methods
- Evaluated student work and collaborated with instructional staff to refine course policies

TPX Product Experience Intern

May 2022 – July 2022

Comcast

Denver, CO

- Designed an internal data management site and updated UI for software engineering teams
- Helped troubleshoot GitHub and terminal issues across company devices

PROJECTS

32-Bit Pipelined Processor + Video Game

- Designed and verified a 32-bit pipelined processor in Verilog implementing hazard detection, forwarding, carry-lookahead arithmetic, and Modified Booth multiplication
- Debugged processor using GTKWave, Vivado, and Python before FPGA deployment to support a custom MIPS video game

Audio Note Recognition Using Fast Fourier Transform

- Developed a MATLAB application using FFT techniques to identify musical notes from complex audio recordings

SKILLS

Technical Skills: Verilog, PSpice, KiCad, PCB Design, AutoCAD, Python, C, MATLAB, Vivado, Microsoft Office, Jira, Trello

Activities: Duke Tour Guides, Theta Tau, DukeEngage, Hoops for Troops